

# PortFlow

AI-assisted congestion prediction and port operations optimization

PortFlow | Innovexa | Bobathon submission

# Problem

Shift supervisors reconcile vessel ETAs, berth limits, yard fill, and disruptions across fragmented

# Solution

Predict delay -> schedule compatible berths -> recommend alternate ports -> brief the supervisor

# Architecture

FastAPI contracts feed a React Gantt timeline and hotspot heatmap. XGBoost predicts; OR-Tools

# Impact

Explainable operations decisions, visible reasoning traces, and a reproducible path from syntheti